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Section _____

**MTH:2310 DISCRETE MATH
MIDTERM 1**

Complete all of the following problems being as detailed as you can be. Remember to fully define new variables/symbols. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit. Do your best to make your solutions organized and written legibly.

Handwriting. (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic or notation errors.

You will be penalized up to 2 points for disorganized or hard-to-read solutions.

Problem 1. (4 points) Define the term *proposition*.

Problem 2. (12 points) State whether the following statements are propositions. **Briefly explain why or why not based on your definition in Problem 1.** If the statement is a proposition, state its truth value.

(a) (3 points) $p = \text{"California is currently one of the 50 states in the United States."}$

(b) (3 points) $q = \text{"The weather is better in California."}$

(c) (3 points) $r = "1+1=3."$

(d) (3 points) $s = "This is the statement provided for Problem 1(e) of Midterm 1 for Section 001 of the MTH:2310 course given in the Spring 2025 semester."$

Problem 3. (6 points) Convert the following statement into a single compound proposition using propositional variables and logical connectors.

Define all variables introduced.

"If I make coffee in the morning and I am running late, I bring my coffee to work."

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Problem 4. (12 points) Let $p = \text{"I am not tired"}$, $q = \text{"I will read a chapter of my book,"}$ and let $r = \text{"I will watch a TV show."}$ Convert the following compound propositions into their corresponding English sentences.

(a) (6 points) $(q \wedge r) \Leftrightarrow p$

(b) (6 points) $p \Rightarrow (q \oplus r)$

Problem 5. (14 points)

(a) (8 points) Complete a truth table for the following two compound propositions.

$$\sim (p \Rightarrow q) \quad \text{and} \quad p \wedge (\sim q)$$

(b) (4 points) Define *logically equivalent propositions* in terms of truth tables.

(c) (2 points) Based on your definition in part (b), are the compound propositions in part (a) logically equivalent?

Problem 6. (8 points) Use a series of logical equivalences to show the following logical equivalence.

The conditional-disjunctive equivalence may be helpful: $P \Rightarrow Q \equiv \sim P \vee Q$.

$$\left((p \Rightarrow q) \Rightarrow r \right) \equiv (\sim p \vee r) \wedge (q \vee r)$$

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Problem 7. (12 points) The following compound proposition is satisfiable. Find an assignment of truth values which satisfy the statement **and show that it satisfies the statement.**

$$p \wedge (q \vee r) \wedge (\sim q)$$

Problem 8. (12 points) Define the terms *tautology*, *contingency*, and *contradiction*.

Problem 9. (16 points) Convert the following statements into a single compound proposition using predicates, quantifiers, and logical connectors. Define all symbols introduced. **Clearly state domains for propositional functions.**

(a) (8 points) "For all natural numbers n , $1 + 3 + 5 + \cdots + 2n - 1 = n^2$."

(b) (8 points) "There is exactly one student in this room who can be the first to turn in their exam."